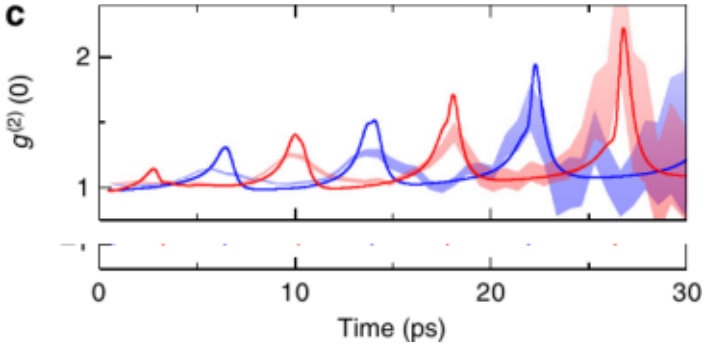
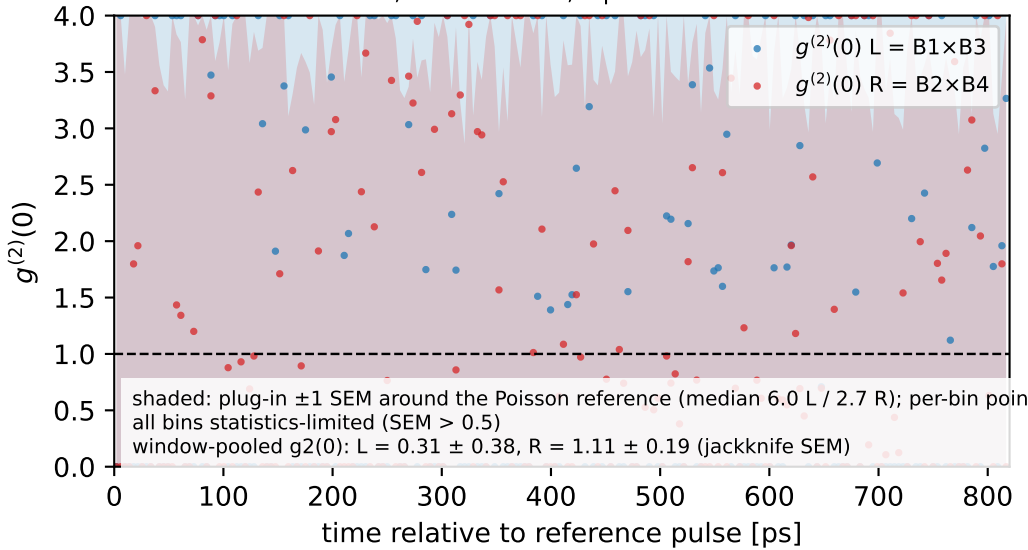
Adiyatullin et al. 2017 (Nat. Commun. 8:1329), Fig. 2(c)




This work: 26-08-04 HBT, 19992 frames, 4 px bins




Left: original Fig. 2(c) of Adiyatullin et al. 2017 (Nat. Commun. 8:1329)— time-resolved zero-delay second-order correlation g(2)(0)(t) of the left (blue) and right (red) mesa; shaded areas are ±1 s.e.m.; the light statistics switches between Poissonian and bunched in phase with the Josephson oscillations, the two mesas in counterphase. Right: this work— same estimator (Suppl. Note 1) on the 26-08-04 HBT batch (19992 single-sweep frames, CFD50 time-zero, measured-background subtraction, 4 px bins). The per-bin regime is statistics-limited: ~50% of detected events are residual detector-noise clusters and the per-bin occupancy is ~1e-3 events/frame, so per-bin g2 is a heavy-tailed statistic. Points are per-bin estimates (clipped to [0, 4]); shaded bands are the plug-in ±1 s.e.m. resolution around the Poisson reference (Monte-Carlo validated), not error bars on the points. Outcome: no significant deviation from Poisson— window-pooled g2(0) = 0.31 ± 0.38 (L), 1.11 ± 0.19 (R) (jackknife s.e.m.); oscillation scan (3-200 ps, reference-lo (p = 0.15); 95% upper limits on the g2-1 amplitude: median 0.81, 1.09 at 9.84ps, 0.73 at 37.2ps, 0.86 at 41.8ps. With 20k frames only the upper part of the paper's amplit range (0.2-1.2) is resolvable; ~3e5 frames would be needed for 95% sensitivity to |A| = 0.2 (G3 power gate, docs/26-08-04-fig2c.md).